

Reliance - TIRA Catalog Enrichment Metric Methodology

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Catalog Enrichment Framework

1. Background

The Catalog Enrichment framework had the general goal of creating a framework to enrich attributes for various channels in Reliance Retail's catalog. Accordingly, the customer and Google team agreed that enriching attributes for Tira beauty would be the quickest way to demonstrate values of this framework. Phase 1 of this framework was developed and implemented for the 80-100 products across 8 to 10 categories within the Tira catalog. The project only used item images with some attributes like L1, L2, L3 category as input, fed them into a Gemini Model to enrich attributes.

During discussions the customer and Google teams agreed that this solution needs to be in the form of a framework. Accordingly Google team has made several decisions of not adding brand specific prompts and logic. We also made this framework stateless so as to avoid technical debt of multiple systems storing copies of the same data. This gives this framework an ability to be used for other channels with minimal development efforts. The downside is that metrics like accuracy etc can be adversely affected for the pilot phase. Once implemented for specific channels, specific prompting and logic can be added to improve performance for that specific channels

2. Metrics Measurement

2.1 Agreed Metrics

Google, Fynd and TIRA Catalog Cloud team met on a series of calls to align on what metrics can be calculated by the Google team. These metrics should have clear calculator logic so they can be calculated programmatically and at scale. Based on the discussions below metrics were agreed upon - [Document Link](#)

Sr No	Metric	Definition
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1	Processing Speed	Time required to enrich SKU (mins)
2	Throughput/Volume	Total SKUs processed per day/week/month manually
3	Attribute Accuracy	Enriched Attributes Vs Ground Truth
4	Data Consistency	Consistency of datatype and LOVs
5	Adaptability	Time and effort required to handle new (SKU) manually

Measurement / unit of measure for each metrics

2.2 Metric Calculations

2.2.1 Processing Speed

This refers to time required to manually enrich a single SKU. It's being measured in minutes. Current values stands around 15-20 mins per SKU for manuall enrichment

2.2.2 Throughput/Volume

This refers to the number of SKUs processed per day/week/month manually. This is measured in absolute numbers.

2.2.3 Attribute Accuracy

This metric measures the correctness of GenAI Enriched Attributes Vs Ground Truth. This is measured in three categories as below

Exact match or similar : Ground truth and enriched value both have same data

Partial Match : AI response has correct values but missing some values from ground truth. This is specifically useful for mutli-select attributes

No Match : AI response is missing all values in ground truth

Below are some observation on sample analysis for the **No Match :**

- Among the `**No match`** cases , it majorly consists of **preference,concern,safety precaution and color hex and description** section.

- For major cases of mismatch in **safety precautions, preference and concern** section is either due to data not present in the image and extracted from other sources.
- For other cases the LLM is evaluating between **description** from paragraph style(in ground truth) to description generated in the dictionary sectional wise, which has similar text but different format and hence marked as a `**No Match**`. These cases can be handled with prompt tuning as well as ground truth format changes.
- The **color hex** attributes are in some cases near to color shade but not similar and can be further improved with a list of values, if possible and prompt tuning.
- The **preference** and **concern** section currently are brand driven but with additional business rule specific generation and prompt tuning.

Sample calculator table below

Ground Truth (attribute + data type)	Enriched Value	Verdict	Explanation
Primary_colour : Gold	Primary_colour : Gold	Exact match or similar	Ground truth and enriched value both have same data that is 'Gold'
Concern : Pores, Dullness, Blemishes	Concern : Dullness	Partial Match	AI response only mentions Dullness, missing Pores and Blemishes
Benefits : Lightweight, Moisturising, Nourishing, Smoothing	Benefits : Sun Protection	No Match	AI response is missing all of the benefits

Below are two key recommendations for Accuracy calculations

1. Reduce number of attributes in the request based on category

All attributes may not be applicable to all L1 categories. Mandatory [attributes mapping](#) is already available to Reliance Team. We recommend that attribute mapping be done at L1 level on Catalog Cloud and only applicable attributes be sent with enrichment calls.

This will have following benefit

- Improve accuracy since model has fewer attributes to work with
- Reduce latency of service

2. Reduce Enum values for attributes in the request based on category

All attributes may not be applicable to all L1 categories. Similarly List of Values(LOVs) are not applicable to all L1 categories . A list of [applicable LOVs](#) is already available to Reliance Team.

We recommend that this mapping be used to reduce the number of Enum values sent with enrichment calls.

This will have following benefit

- Improve accuracy since model has fewer attributes to work with
- Reduce latency of service

2.2.4 Data Consistency

Initial discussion with the customer revealed that earlier attempts resulted in incorrect formats of enriched attributes. Hence Google team built logic to validate if the enriched attribute has correct datatype as well as valid LOV

This metric measures if

- The enriched response is correct datatype
- The enriched response value falls with LOV if applicable

Below tables gives scenarios of `spf` as attributes

Input (attribute + data type)	Enriched Value	Verdict	Explanation
spf : Type - string LOVs ["Below 15","Between 15-30","Above 30"]	spf : Below 15	passed	Enriched value is of type string and falls within LOV
spf : Type - string LOVs ["Below 15","Between 15-30","Above 30"]	spf : 15	failed	Enriched value is of type integer and also does not fall within LOV
spf : Type - string LOVs ["Below 15","Between 15-30","Above 30"]	spf : Above 50	failed	Enriched value is of type string but does not fall within LOV
spf : Type - string LOVs ["Below 15","Between 15-30","Above 30"]	spf : ""	failed	Enriched value is of type string but does not fall within LOV since it's blank

2.3 Evaluation on Golden Data Set

Groundtruth dataset identified by Reliance team. The enrichment framework was run on this dataset to arrive at initial values for [agreed metrics](#). The Google team conducted meetings with the Tira Catalog team to identify attributes that are manually enriched. The number of attributes identified for enrichment was outcome of these discussions

The dataset was chosen to be representative sample of catalog with below distribution

Unique SKU's - 85

Product Category Coverage - 9 L1 category

Description	Existing (Manual Update)	GenAI Solution
Total Attributes Identified for Enrichment	2030	2030
Number of Attributes Filled	745	1085
Uplift		45.64%
Enrichment Time (single SKU)	10-15 mins	~ 14 secs
Enrichment Time (entire set)	1000-1500 mins	20 mins
Uplift		50-75X
Accuracy	NA*	0.79
Data consistency	NA*	0.96
Request Fill Rate	NA*	0.51*

NA* : No clear Indication or tangible method to calculate this when it's being done manually

Fill rate considerations

The current framework allows all 25 attributes to be passed in the request body for all L1 categories. However not all 25 attributes are applicable for all categories as given [here](#). Currently this attribute-category level mapping is not available in Catalog Cloud.

For clarity, we can look at following flavours of fill rate

- Request fill rate :
 - Calculated for every request right now
 - Calculated based on number of attributes sent in request body and responses generated by Gemini Model
- Actual Fill rate :
 - Calculated at Sample set level (Golden Set , Set of 300 SKUs and so on)

- Calculated based on ground truth availability

If we consider actual fill rate, i.e. attributes enriched only where ground truth was available, the number is currently at around 95% as calculated here . Once attribute-category mapping is available and [recommended attributes in request is reduced](#), the Request fill rate will go drastically up and be close to Actual Fill rate

The enrichment of the attributes depends on the image and L1 category of the product but for GenAI based enrichment we are sharing all the list of permissible attributes to the model and currently the model is prompted to reply empty string or default empty value for all those attributes which are currently not available in the image or not enrichable.

We are marking them as “**NA status**” for the fill rate metrics calculation.

We used below validation status for Fill rate calculation

Status	Meaning
Passed	Status where the enriched data meets the data type and LOV requirement
Failed	Status where the enriched data does not meet the data type and LOV requirement
NA	Status where the Genai model shares an empty response as it is not available in the source images.

For example

For product “**SUGAR Cosmetics Bling Leader Illuminating Sunscreen**” the below are the split for enrichment status for all the 25 attributes to be enriched when compared to ground truth.

passed_status	na_status	failed_status
12	12	1

So the fill rate calculator for the same is

None

$\text{fill_rate} = \frac{\text{passed_status} + \text{failed_status}}{\text{passed_status} + \text{failed_status} + \text{na_status}}$

Fill rate = $12 + 1 / (12 + 12 + 1) = \mathbf{0.52}$ for the product.

2.4 Evaluation on Top 3000 SKUs by popularity from TIRA Catalog

Tira catalog team provided 3000 SKUs spread across three top selling L1 categories on Tira website. Google Team ran enrichment and calculated the below metrics

The dataset was chosen to be representative sample of catalog with below distribution

Unique SKU's - 2987

Product Category Coverage - 3 L1 category

Description	Existing (Manual Update)	GenAI Solution
Total Attributes Identified for Enrichment	74675	74675
Number of Attributes Filled	22891	38802
Uplift		69.49%
Enrichment Time (single SKU)	10-15 mins	~14 secs
Enrichment Time (entire set)	~498 -746 hrs (~23-31 days)	141 mins
Uplift		207 -317X
Accuracy	NA*	0.787
Data consistency	NA*	0.94
Request Fill Rate	NA*	0.51

3. Business Metrics and Impact

This section translates the impact of the observed metrics on some of the business parameters not necessarily measurable immediately

3.1 Throughput

Currently 15-20 (person-mins) are spent to enrich per SKUs manual. This translates to 1500-2000 person-mins for 100 SKUs daily. With Gemini based enrichment framework this can be brought down to ~30 sec per SKU i.e 30 mins per day.

AI based frameworks can also scale horizontally meaning enrichment calls for all 100 SKUs can be made simultaneously and independent of each other further increasing throughput.

With current batch script approach, enrichment of 100 SKUs took around 20 mins and 3000 SKUs took around 141 mins

3.2 Cost

Manual efforts have direct and indirect costs associated with it. Managing the human resource pool as additional overheads such as supervision costs and so on. Increased throughput can reduce costs on both these fronts.

Enrichment time reducing from 15 mins to 30 sec translates to ~96% cost savings directly on human resource costs. Additionally lesser manual efforts meaning smaller resource pool and proportionally lesser overhead costs

3.3 Time to Market:

AI based catalog enrichment framework can reduce time taken to onboard new products manually . This means faster availability of products and could mean significant competitive advantage in case of high visibility events such as seasonal sales and so on

3.4 User Satisfaction

A well enriched catalog can dramatically improve user experience by improving functionality of filtering. Additionally enriched attributes can be consumed by search and recommendation systems thereby influencing basket size.

4. Engagement Summary (UC6: Catalogue Enrichment)

4.1 Problem Statement

The Catalog Enrichment framework was primarily developed to create a generalized structure for enriching attributes for various channels in Reliance Retail's catalog. The initial focus was enriching attributes for Tira beauty, which was decided by both Reliance and Google teams to showcase the value of this framework. During Phase 1, the framework was developed and implemented for 80-100 products across 9 L1 categories within the Tira catalog, using product images as the only source of information. Additionally, an automated LLM-based evaluator mechanism was included to simplify the evaluation process and act as an initial enrichment validator. This phase successfully enriched 25 attributes from the TIRA Catalogue and the framework was also tested for scalability on 3K product catalogs of TIRA..

4.2 Google Solution (How we helped)

- Developed framework for catalog attribute enrichment which is consumable as a stateless API allowing for quick and seamless integration across systems
- Implemented a modularized approach which can be extended/re-used with minimum effort to multiple verticals and domains.
 - F/w allows for defining specific attributes to be generated relevant to a product category
 - F/w allows defining user prompts specific to attributes and hence adapts well to other verticals
 - F/w support multiple output formats for attribute generation - dictionary, Integer, float, string and array
- Customizable format generation for specified attributes. F/w allows to define formats at attribute level which is referred to during attribute generation
- Ensured reduction in error rate (vs manual process) by providing robust checks for following

- Type Validation for each generated attribute
- Values in-range check for prescribed attributes with a list of valid values
- Robust validation and evaluation modules for the below
 - Supporting Multi select with list of valid values.
 - Automated LLM based validation for generated outcomes reducing human validations efforts & time for the entire catalog

4.3 Additional Modules Developed

- Batch Script - The Google team has provided a script to call the enrichment framework in bulk. This will enable the customer engg team to quickly experiment and iterate in next phases.
- LLM based Evaluation Script - The Google team also provided a LLM based evaluation script. This will enable the customer engg team to evaluate future larger experiments with minimal manual intervention. This script can be re-used to calculate agreed metrics at scale for other channels
- Additional attribute generation - Information needed to enrich all attributes may not be part of images for all products. Demonstrated generation of additional attributes via identified data sources. Once shared these can be ingested and used by Gemini to generate additional attributes. Gemini with Search Grounding and Programmable Search engine can potentially help pull this information from product websites and other sources available on the internet. This will further improve attribute fill rate. This is currently being discussed with Product team

4.4 Impact

- Analyzed TIRA beauty catalog with 94K unique SKU's across 9 L1 categories of product to finalize the potential attributes for enrichment using image as the input data
- Processed **3100 unique SKU's** across all product categories for the TIRA catalogues
- Enriched **75000 attributes** with the product catalog enrichment pipeline. Uplift of **69.5%**
- Attribute Enrichment time reduction minutes , massive **100x reduction** from the current manual process (calculated on a bundle of 100 SKU's)

- Calculated Data Consistency - 96%
- Calculated Accuracy - 79% (Measured only on attributes with Groundtruth which is 31 % of total attributes . Please refer to section 2.2.3 on calculations. Overall > 85%)
- Current Users - Catalog cloud team
- Projected Users - Catalog Cloud team and Tira Enrichment team